## DOCUMENT CONTROL

| | |
|---|---|
| **Document ID** | EAAP-DEF-001 |
| **Title** | Enterprise AI Architecture Pattern: Framework Definitions & Master Reference |
| **Version** | v1.0 |
| **Status** | Release |
| **Classification** | Framework Core / Controlled: Master Reference |
| **Layer** | Layer 0: Framework Foundation |
| **Parent** | None: This is the master root document |
| **Owner** | John Goulden |
| **Date** | 23.04.2026 |
| **Contact** | research@johngoulden.de |
| **Child Documents** | EAAP-CORE-001, EAAP-ARCH-001 through EAAP-ARCH-006 |

### Version History

| Version / Date | Author | Description |
|---|---|---|
| **v1.0 \| 23.04.2026** | John Goulden | Initial release. Master parent document for the EAAP document hierarchy. Resolves all child document parent references. |

# Table of Contents

## 1. PURPOSE AND AUTHORITY

This document is the master root of the Enterprise AI Architecture Pattern (EAAP) document hierarchy. It establishes the foundational definitions, framework identity, scope boundaries, document register, and governing rules that all child documents must conform to.

All EAAP documents reference this document as their ultimate authority. In any conflict between this document and a child document, this document prevails.

> *EAAP-DEF-001 is the parent of all EAAP documents. No EAAP document exists outside this hierarchy. No EAAP document may contradict the definitions or governing rules established here.*

## 2. FRAMEWORK IDENTITY

| | |
|---|---|
| **Framework name** | Enterprise AI Architecture Pattern |
| **Abbreviation** | EAAP |
| **Author and owner** | John Goulden |
| **Origin date** | March 2026 |
| **Current version** | v1.0 |
| **Framework classification** | Proprietary architectural pattern and intellectual property |
| **Deployment reference** | EAAP Enterprise AI Platform (reference implementation) |
| **Governing law** | Federal Republic of Germany |

## 3. SCOPE OF EAAP

### 3.1 What EAAP includes

- **Abstract architectural principles** The design pattern for modular, governable, vendor-independent enterprise AI systems.
- **Orchestration framework** The principles and decision model governing how multiple AI domain agents are coordinated.
- **Domain agent model** The structural specification all domain agents must conform to.
- **Governance architecture** The structural controls for traceability, auditability, escalation, and human oversight.
- **Knowledge integration model** The principles governing how knowledge sources ground agent reasoning.
- **Deployment neutrality principles** The requirements for platform-agnostic, air-gappable deployment.
- **IP boundary definitions** The precise scope of what constitutes EAAP intellectual property.

### 3.2 What EAAP excludes

- **Individual domain agent implementations** Specific system prompts and agent operating content are implementation artefacts, not framework definitions.
- **Customer-specific deployments** Any deployment of EAAP for a specific organisation is outside the abstract framework.
- **Enterprise Orchestration Layer** Higher-level inter-agent coordination, meta-agent control logic, and enterprise-wide orchestration remain separate IP unless expressly included by written amendment.
- **Executable software components** EAAP is an architectural pattern, not a software product.
- **Third-party products** References to specific vendors (Nutanix, VMware, etc.) are implementation examples, not framework components.

## 4. CORE DEFINITIONS

The following definitions apply across all EAAP documents. Where a child document uses any of these terms, the definition below governs.

| Term | Definition |
|---|---|
| EAAP | Enterprise AI Architecture Pattern. The abstract, proprietary design pattern defined and owned by John Goulden, consisting of the framework documents listed in Section 6. |
| Domain Agent | A bounded AI reasoning component dedicated to one technical, operational, legal, or business domain. Governed by EAAP-ARCH-001. |
| Orchestrator | The supervisory reasoning layer that coordinates multiple domain agents, resolves conflicts, assigns confidence, and enforces escalation. Governed by EAAP-ARCH-002 and EAAP-ARCH-003. |
| Knowledge Layer | The structured knowledge sources that ground domain agent and orchestrator reasoning. Evidence-classified, version-controlled, and locally hosted. |
| Governance Layer | The structural controls applied across all EAAP layers ensuring traceability, auditability, human oversight, and separation of recommendation from execution. |
| Automation Gateway | The sole component permitted to interact with target infrastructure systems. Implements plan-approve-execute workflow with deny-by-default access control. |
| Evidence Class | The classification of the source supporting an agent or orchestrator output: Vendor-Supported, Framework-Supported, Best Practice, Heuristic, or Assumption. |
| Execution Boundary | The architectural boundary separating AI reasoning from enterprise action. The agent boundary ends at recommendation. Execution requires explicit human approval. |
| Confidence Level | The assessed reliability of an output: High, Medium, Low, or Unknown. Governed by the confidence model in EAAP-ARCH-003. |
| Action Risk Tier | The risk classification of a recommended action: Tier 0 (read-only), Tier 1 (low-risk reversible), Tier 2 (disruptive), Tier 3 (destructive). |
| Data Sufficiency Gate | The mandatory check before domain agent invocation that confirms sufficient input data is available to reason reliably. |
| Conditional Override | The governance halt triggered when confidence is Low, data is insufficient, or domain conflict is unresolved. No recommendation is issued until the condition is resolved. |
| Framework Asset | Any document, prompt specification, reasoning workflow, architectural pattern, or design methodology that forms part of the EAAP intellectual property. |
| Implementation | A specific deployment of EAAP-compatible architecture for a defined environment. Distinct from the abstract framework. |
| Reference Implementation | The EAAP Enterprise AI Platform. The sole current reference implementation of the EAAP framework. |
| Productive Use | Any use of the EAAP framework beyond evaluation, including operational deployment, commercial licensing, or integration into products or services. |

# 5. ARCHITECTURAL LAYERS

## 5.1 Layer model

EAAP is organised into four architectural layers. Each layer has a defined function and interacts with adjacent layers only through controlled interfaces.

| Layer | Number | Description | Documents |
|---|---|---|---|
| **Framework Foundation** | Layer 0 | Master definitions, scope, and governing rules. | EAAP-DEF-001 |
| **Architecture Pattern** | Layer 3 | Abstract architectural principles independent of any implementation. | EAAP-CORE-001 |
| **Architecture Design** | Layer 2 | Formal architecture models, decision models, and functional specifications. | EAAP-ARCH-001 through EAAP-ARCH-004, EAAP-ARCH-006 |
| **Implementation** | Layer 1 | Specific deployment designs, technology choices, and production specifications. | EAAP-ARCH-003A, EAAP-ARCH-005 |

## 5.2 Layer descriptions

### Layer 0: Framework Foundation

This document. Defines the framework identity, master definitions, document hierarchy, and governing rules. No other document supersedes Layer 0.

### Layer 3: Architecture Pattern

Defines abstract architectural principles independent of any vendor, infrastructure platform, LLM, or deployment environment. EAAP-CORE-001 lives at this layer. It cannot reference specific products or implementations.

### Layer 2: Architecture Design

Defines formal architecture models, decision frameworks, and functional specifications derived from Layer 3 principles. Documents at this layer define how EAAP works but remain platform-neutral.

### Layer 1: Implementation

Defines specific deployment designs with named technologies, version numbers, and environment-specific configurations. Documents at this layer describe a specific realisation of the EAAP architecture.

# 6. DOCUMENT HIERARCHY

## 6.1 Document register

| Document ID | Title | Layer | Status | Parent |
| --- | --- | --- | --- | --- |
| **EAAP-DEF-001** | Framework Definitions & Master Reference | Layer 0 | Release | None: Root |
| **EAAP-CORE-001** | Core Architecture Definition | Layer 3 | Release | EAAP-DEF-001 |
| **EAAP-ARCH-001** | Domain Agent Model | Layer 2 | Release | EAAP-DEF-001 |
| **EAAP-ARCH-002** | Orchestration Logic | Layer 2 | Release | EAAP-DEF-001 |
| **EAAP-ARCH-002A** | Orchestration Prompt Control | Layer 2 | Release | EAAP-ARCH-002 |
| **EAAP-ARCH-003** | Orchestration Decision Model | Layer 2 | Release | EAAP-DEF-001 |
| **EAAP-ARCH-003A** | Orchestration Decision Model: Production Design | Layer 1 | Production Design | EAAP-ARCH-003 |
| **EAAP-ARCH-004** | Functional Architecture Model | Layer 2 | Release | EAAP-DEF-001 |
| **EAAP-ARCH-005** | Functional Architecture HLD | Layer 1 | Draft | EAAP-ARCH-004 |
| **EAAP-ARCH-006** | Deployment Model: Platform-Agnostic | Layer 2 | Release | EAAP-DEF-001 |

## 6.2 Parent-child rules

- **Every EAAP document must declare a parent document in its Document Control section.**
- **No document may contradict a definition or governing rule established in a parent document.**
- **A child document may extend and specialise a parent document's scope but may not reduce it.**
- **Version increments of a parent document require review of all child documents for consistency.**
- **Documents at Layer 1 may reference specific technologies. Documents at Layer 2 and above must remain platform-neutral.**

# 7. VERSIONING AND CHANGE CONTROL

| Version increment | Trigger | Approval required |
| --- | --- | --- |
| **Patch (x.x.N)** | Corrections to existing content. No scope change. | Owner review. |
| **Minor (x.N.0)** | Addition of new sections or definitions. No structural change. | Owner review and version history update. |

| Version increment | Trigger | Approval required |
|---|---|---|
| **Major (N.0.0)** | Structural change, scope change, or redefinition of core terms. | Owner review, child document impact assessment, and version history update. |

*A major version increment of EAAP-DEF-001 requires review and update of all child documents. Child documents must not be published referencing a superseded version of this document without documented justification.*

## 8. INTELLECTUAL PROPERTY DECLARATION

The EAAP framework, including all documents in the register defined in Section 6, constitutes pre-existing intellectual property developed independently by John Goulden.

This IP includes but is not limited to:

- **All framework definitions, principles, and governing rules defined in this document.**
- **All architectural models, decision frameworks, and functional specifications in Layer 2 documents.**
- **All orchestration logic, conflict resolution models, and confidence models.**
- **All domain agent structural specifications and governance patterns.**
- **All deployment models, security architectures, and integration patterns.**

*Separately and expressly excluded from the present framework: any future-developed Enterprise Orchestration Layer including inter-agent reasoning coordination, supervisory prompt routing, meta-agent control logic, enterprise-wide domain interaction frameworks, and centralized reasoning arbitration mechanisms. These constitute separate intellectual property in development.*

## 9. GOVERNING RULES

The following rules apply to the entire EAAP framework and all documents within it. These rules are non-negotiable and cannot be overridden by child documents.

| Rule | Statement |
|---|---|
| GR-001 | All EAAP documents must reference EAAP-DEF-001 as the ultimate authority. |
| GR-002 | No domain agent may be deployed under EAAP without conforming to EAAP-ARCH-001. |
| GR-003 | No multi-agent orchestration may bypass the logic defined in EAAP-ARCH-002 and EAAP-ARCH-003. |
| GR-004 | No Tier 2 or Tier 3 action may be executed without explicit human approval through the Automation Gateway. |
| GR-005 | All outputs must carry evidence classification, confidence level, and risk tier. |
| GR-006 | Governance is structural. It may not be made optional or bypassed by configuration. |
| GR-007 | The execution boundary is absolute. Agents advise. Humans decide. Gateways execute. |
| GR-008 | Low or Unknown confidence blocks all recommendations. Data collection is required before re-invocation. |
| GR-009 | Conflicts between domain agents must be surfaced explicitly. Silent conflict resolution is prohibited. |
| GR-010 | No document outside the register in Section 6.1 may claim to be part of the EAAP framework without written amendment to this document. |
| GR-011 | EAAP governance accountability is bounded to the Deployment Risk domain. AI Model Risk accountability rests with the AI vendor. AI Operational Risk accountability rests with the operating entity. These three accountability domains must be formally assigned, documented, and must not be conflated. An AI deployment may not proceed until all three accountability domains have named owners, the Model Risk Assessment Gate (TBM-D-001) has been completed and signed, and the gate record has been registered in the EAAP audit trail. |

## GLOSSARY

All terms defined in Section 4 constitute the authoritative EAAP glossary. Child documents may add domain-specific terms but may not redefine terms established here.

research@johngoulden.de